

QUANTIFYING THE INTERNAL VALIDITY OF WEIGHTED ESTIMANDS

A discussion of Poirier and Słoczyński (2026)

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Class of weighted estimands: (assume $W_0 = 1$ w. probability 1)

$$\mu(a, \tau_0) = \frac{\mathbb{E}[a(X)\tau_0(X)]}{\mathbb{E}[a(X)]}$$

Includes OLS, 2SLS, IV, TWFE as special cases (different a , different W_0).

Two questions:

1. Can $\mu(a, \tau_0)$ be written as $\mathbb{E}[Y(1) - Y(0) \mid W^* = 1]$ for some subpopulation W^* ?
 - W^* is a subpopulation whose selection is quasi-random
2. How large is $\mathbb{P}(W^* = 1 \mid W_0 = 1)$?
 - What share of the target quantity is identified by the weighted estimand?

Headline answers:

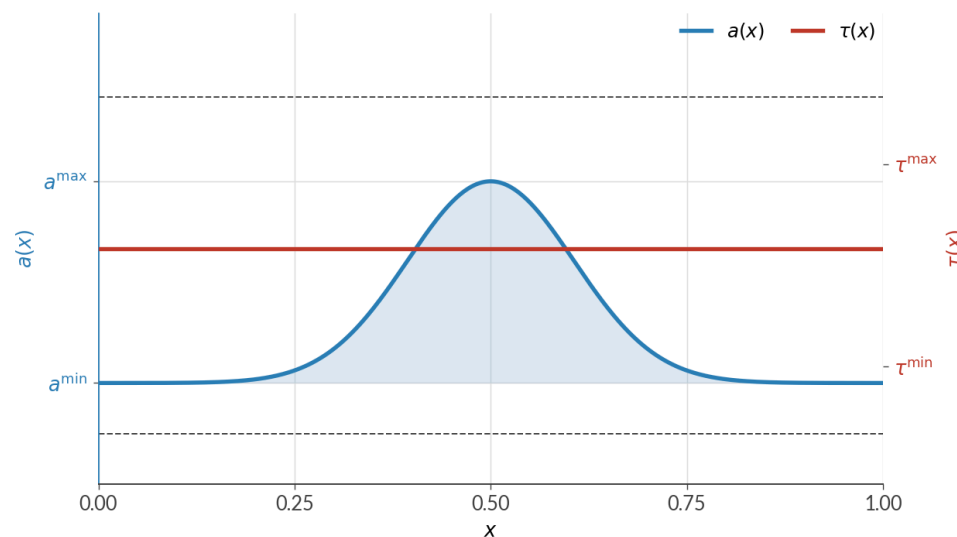
- Uniform-in- τ_0 : need $a(X) \geq 0$; size = $\frac{\mathbb{E}[a(X)]}{a_{\max}}$
- Given τ_0 : need μ_0 in convex hull of CATEs; linear program for size

On $L^2(F_{X|W_0=1})$ with $\langle f, g \rangle = \mathbb{E}[f(X)g(X) \mid W_0 = 1]$:

$$\mu(a, \tau_0) = \frac{\langle a, \tau_0 \rangle}{\langle a, 1 \rangle}$$

$$\mu(a, \tau_0) - \text{ATE} = \frac{\text{cov}(a, \tau_0 \mid W_0 = 1)}{\mathbb{E}[a \mid W_0 = 1]}$$

Existence result asks: is there a group across X such that their average τ_0 equals $\mu(a, \tau_0)$?

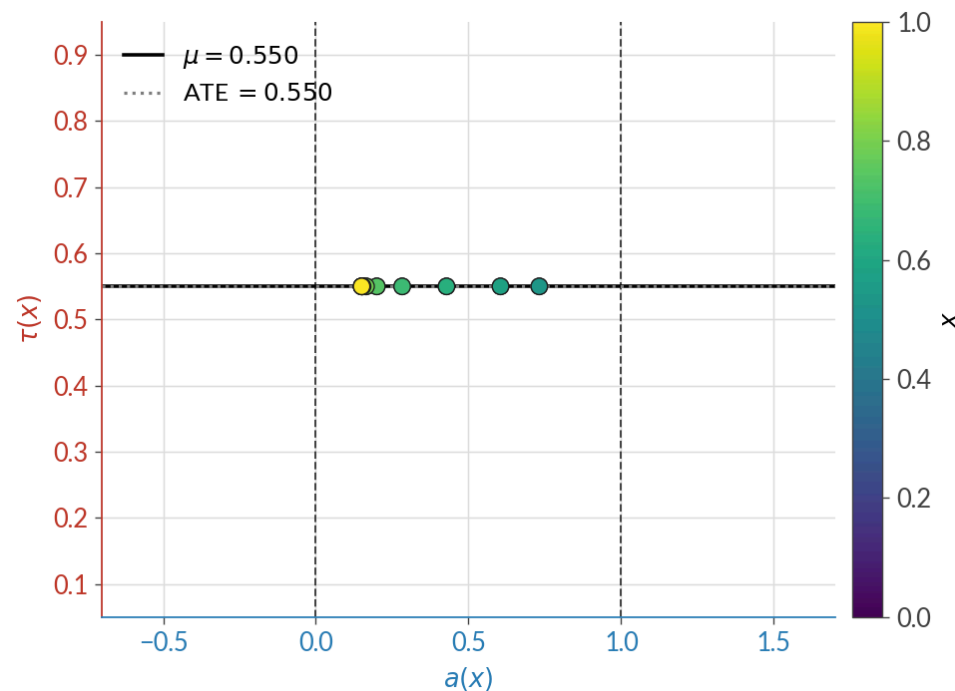


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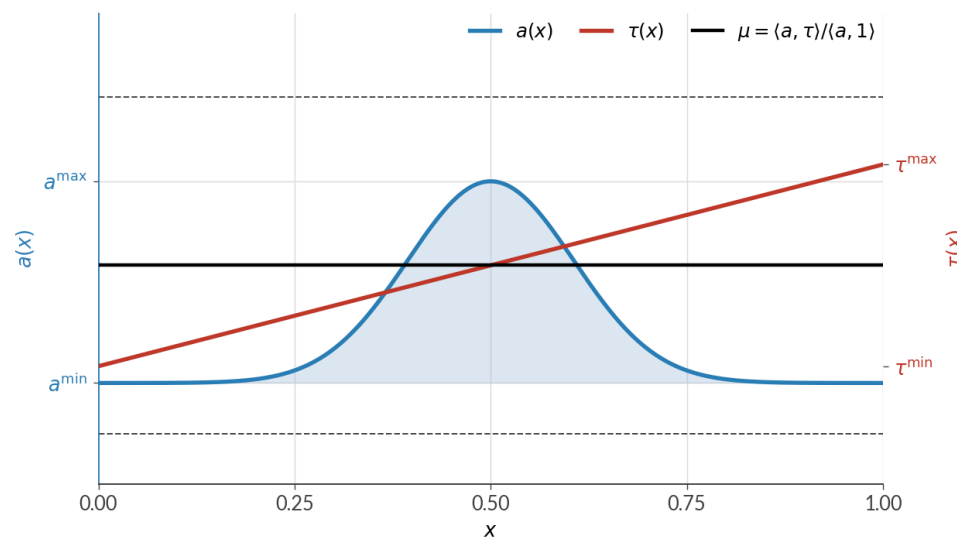


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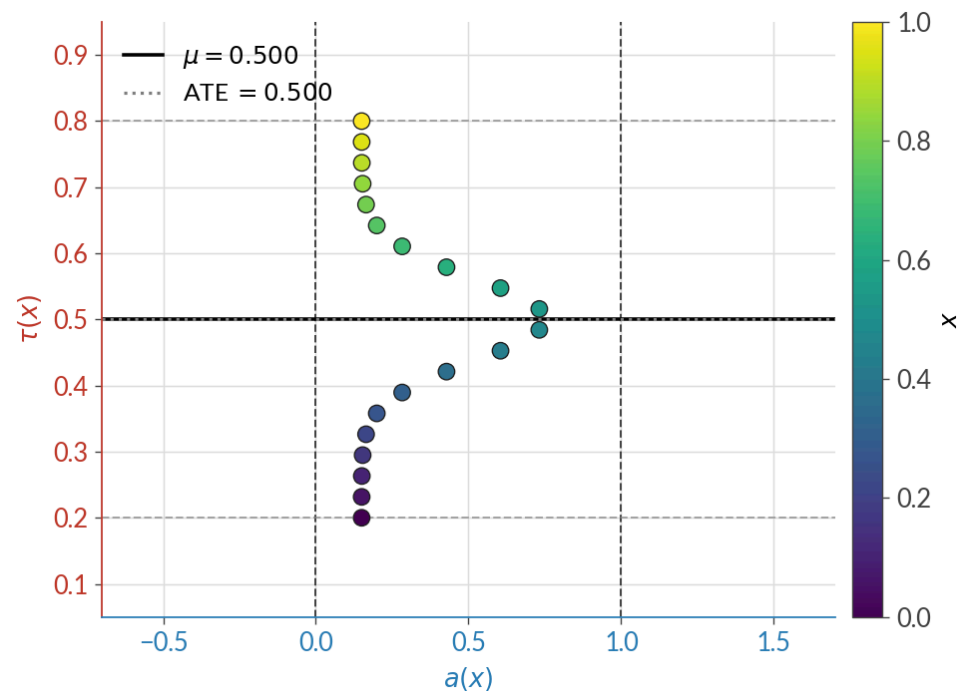


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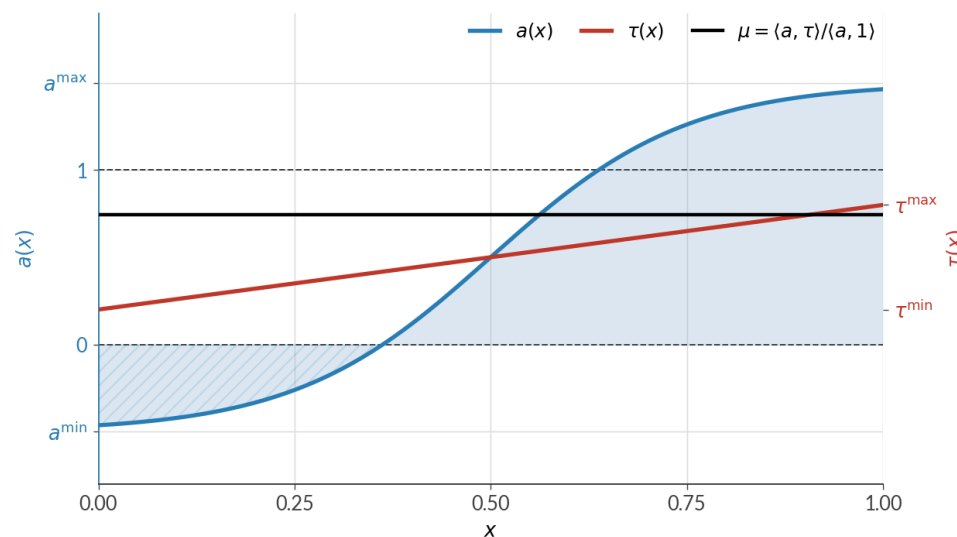


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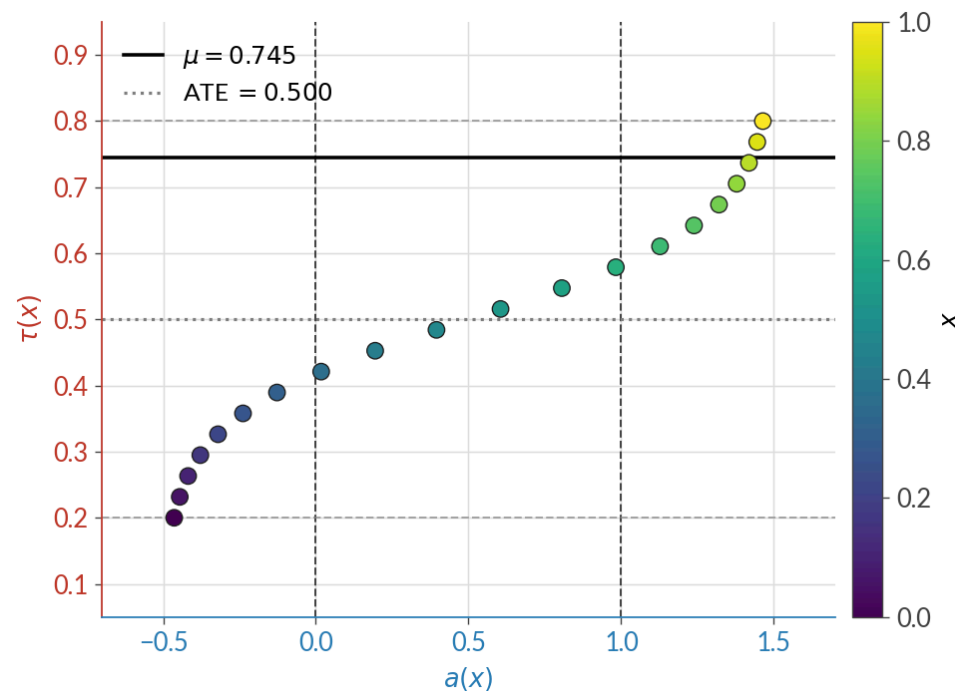


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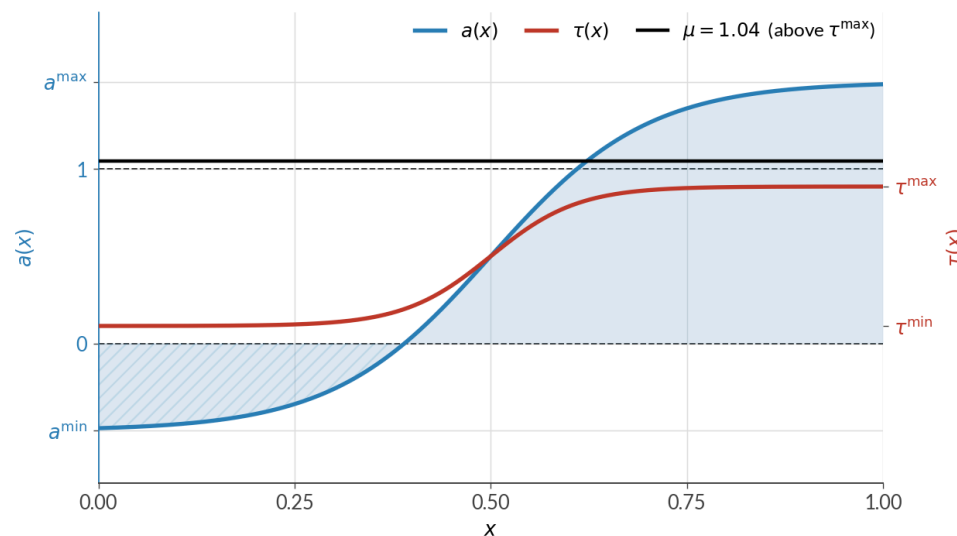


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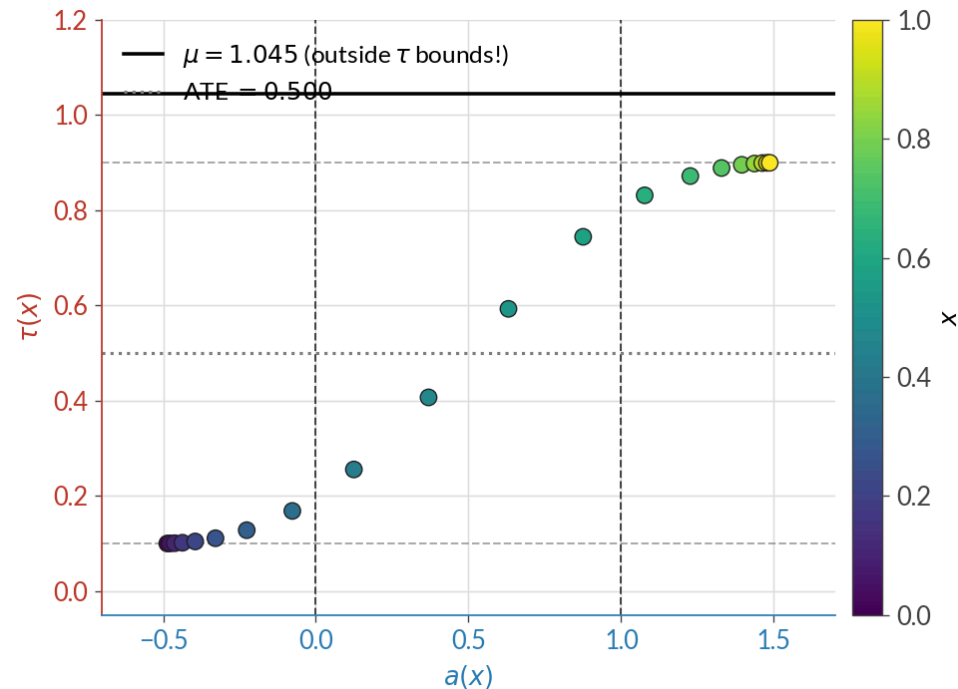


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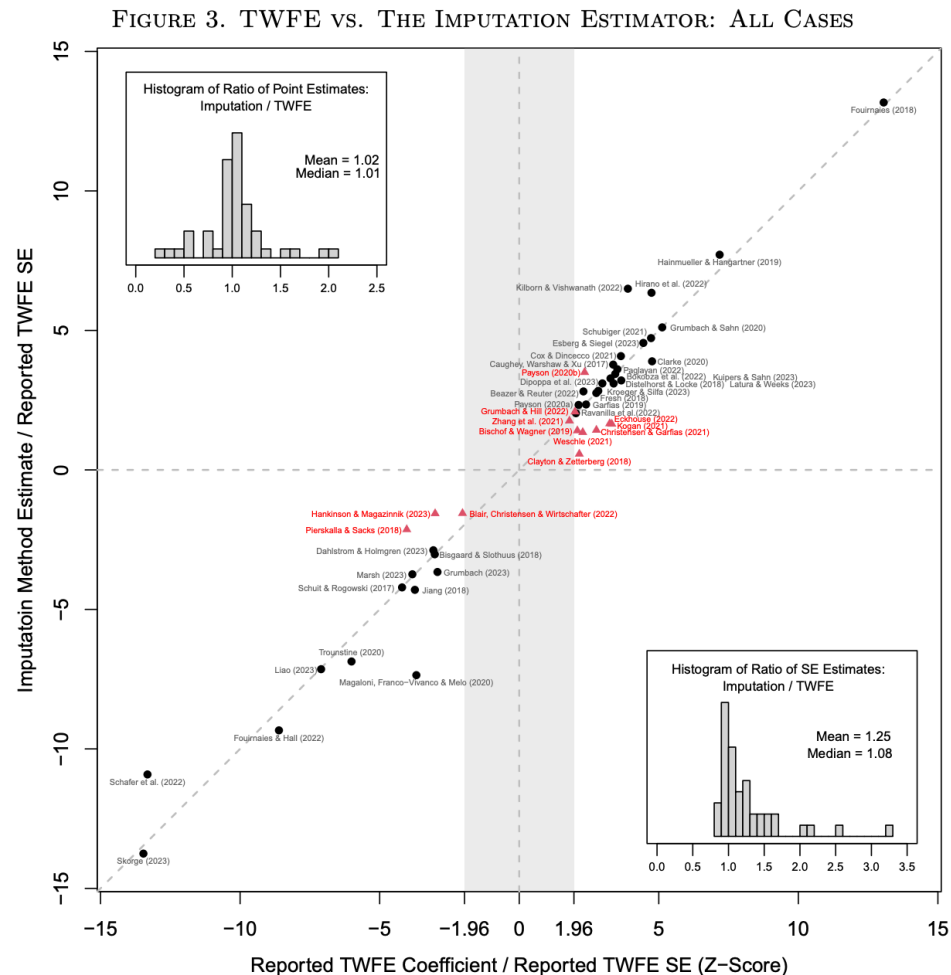
Results are **worst-case** over $\mathcal{T}_{\text{all}}$: all functions with finite second moments.

The worst case is achieved by the adversarial $\tau^-(X) = \mathbb{1}(a(X) < 0)$, τ is spiked exactly where a is large.

- These τ 's may be implausible: discontinuous, atomic, perfectly aligned with weight signs.
- What about considering alternative bounds? Lipschitz smoothness?

Limited evidence that heterogeneity is adversarially aligned with weights.
Heterogeneity is real, but $\text{cov}(a, \tau_0) \approx 0$.

1. Chiu, Lan, Liu and Xu (2026, ASPR) reanalyze 49 DiD TWFE applications and find very limited deviations using alternative estimators, suggesting limited adversarial alignment.
2. In GPHK (2024), we found that contamination bias weights had limited covariance with CATEs in RCTs.



Frame	Object	Useful when
PS-uniform	$\sup_{\tau_0 \in \mathcal{T}_{\text{all}}}$	Adversarial robustness; no information on τ
PS-given-τ_0	Point answer at $\hat{\tau}_0$	You trust the CATE estimate
A potential alternative? Prior on τ_0	Posterior on τ	Structural beliefs about τ : smooth, weakly related to a , etc.

- Paper studies this in a very nice linear example
 - Much weaker restrictions on weights to get “causal” estimates

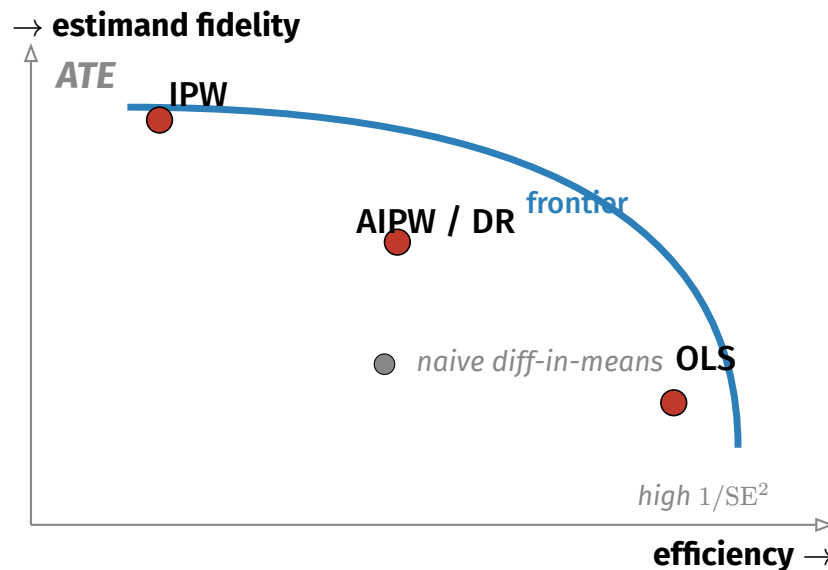
Why do we weight these things so weirdly?

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 - why are you cooking salmon with the dishwasher?
- Folk wisdom: we use weighted estimands because moving down the frontier buys precision — at the cost of estimand fidelity
 - GPHK (2024) “Easiest-to-estimate”
- This estimand fidelity loss is higher depending on the CATE function
 - Empirically, I think people are not worried – should they be?



- $\tau(X)$ is not known in practice – we estimate $\hat{\tau}(X)$
- If $\hat{\tau}(X)$ is very noisy in places, we often target estimands other than the ATE
 - e.g. Crump et al. (2009)
- PS results give a way to ask how representative the proposed weighting is relative to target estimand
 - But, all based on **estimated** $\hat{\tau}(X)$ – how noisy?
- Concerns particularly large when X is high-dimensional
- If we consider a prior distribution over $\mathcal{T}_{\text{all}}$, can we directly target this?

Thanks!

- **Design-based** (stratified RCT, mover designs): $p(X)$ linear in strata indicators **by construction**. PS machinery applies cleanly.
- **Model-based** (observational, selection on covariates): linearity is an **added** approximation assumption. Nonlinearity injects further bias on top.

Implication: PS-uniform's diagnostic power is highest where it's least needed (designs) and lower where empirical concerns are greatest (obs studies).

PS Appendix C: nonstandard limiting distribution for $\hat{\mathbb{P}}$, Fang-Santos bootstrap.

Practical questions for Table 1 (divorce / suicide):

- Uniform $\hat{\mathbb{P}} = 0$, given $\tau_0 \hat{\mathbb{P}} = 0.62$. What are the CIs?
- The 0 vs. > 0 distinction is binary in the paper but a continuous estimation problem. How sensitive?
- Across applications: is the gap between uniform and given τ_0 generally large (as GPHK predicts)? Worth reporting alongside point estimates.